Lab Report

Magnetometry with spins in diamond

PHY405: Condensed Matter Physics Laboratory

Naini Diwan
Roll no.: 23199

Contents

1 Objective **2**

2 Theory **2**

2.1 Ground state of a single NV centre 2

2.2 Four orientations in an ensemble 2

3 Experimental arrangement and conditions **3**

3.1 Set-up 3

3.2 Coil geometry and reference frames 3

3.3 Recorded data 4

4 Data analysis **4**

4.1 Fit model 4

4.2 Noise level 4

5 Part A **5**

5.1 Q1. Components of the set-up and their function 5

5.2 Q2. Influence of the field projection on the number of dips 5

5.3 Q3. Role of the microwave field 5

6 Q4. Current values, Magnetic field and Zeeman analysis **6**

6.1 Coil field formula 6

6.2 Field vector in the diamond frame and NV projections 6

6.3 Case 1: one dip ($\mathbf{B} = 0$) 6

6.4 Case 2: two dips (field along z) – Series B 6

6.5 Case 3: three dips (field along [110]) – Series A 9

6.6 Case 4 and mixed field configurations – Series C 12

6.7 Is there another way to get four dips? 16

7 Discussion **16**

8 Q5. NV centres: insight into physics and applications **17**

9 Conclusions **18**

A Fit parameters **18**

B Python code used for the fits and field calculation **19**

1 Objective

The aim of this experiment is to measure the Zeeman splitting of the electron-spin resonances of an ensemble of nitrogen-vacancy (NV) centres in a (100)-oriented HPHT diamond by optically detected magnetic resonance (ODMR), and to use it to
(i) operate a diamond magnetometer,
(ii) control the direction and magnitude of a magnetic field with two perpendicular coils,
(iii) determine the projection of the applied field on each of the four NV orientations.

2 Theory

2.1 Ground state of a single NV centre

The ground state of the negatively charged NV centre is a spin-1 triplet. With the quantisation axis z along the NV axis and $\hbar = 1$, for small static fields

$$\hat{H}_0 = D\hat{S}_z^2 + \gamma B_z \hat{S}_z, \qquad D \approx 2.870 \text{ GHz}, \quad \gamma = 28 \text{ MHz/mT}, \tag{1}$$

so that $E_{|0\rangle} = 0$ and $E_{|\pm1\rangle} = D \pm \gamma B_z$. Only the field component B_z along the NV axis enters to first order; transverse components shift the levels only at second order, by roughly $(\gamma B_\perp)^2/D \lesssim 1.5$ MHz for the fields used here. Resonant microwaves drive $|0\rangle \to |\mp 1\rangle$ at $f_\mp = D \mp \gamma B_z$, so the two dips of one NV orientation are separated by

$$\Delta\nu = f_+ - f_- = 2\gamma B_z. \tag{2}$$

2.2 Four orientations in an ensemble

In the diamond frame (x, y, z) the four NV axes are $\hat{u}_1 = \frac{1}{\sqrt{3}}(1,1,1)$, $\hat{u}_2 = \frac{1}{\sqrt{3}}(-1,1,-1)$, $\hat{u}_3 = \frac{1}{\sqrt{3}}(1,-1,-1)$, $\hat{u}_4 = \frac{1}{\sqrt{3}}(-1,-1,1)$, & applied $\mathbf{B} = (B_x, B_y, B_z)$ has the projections

$$\begin{aligned} B_1 &= \tfrac{1}{\sqrt{3}}(B_x + B_y + B_z), & B_2 &= \tfrac{1}{\sqrt{3}}(-B_x + B_y - B_z), \\ B_3 &= \tfrac{1}{\sqrt{3}}(B_x - B_y - B_z), & B_4 &= \tfrac{1}{\sqrt{3}}(-B_x - B_y + B_z). \end{aligned} \tag{3}$$

Orientation i contributes a pair of dips at $f_{i\pm} = D \pm \gamma|B_i|$, giving up to eight dips. Dips coincide when projections have equal magnitude.

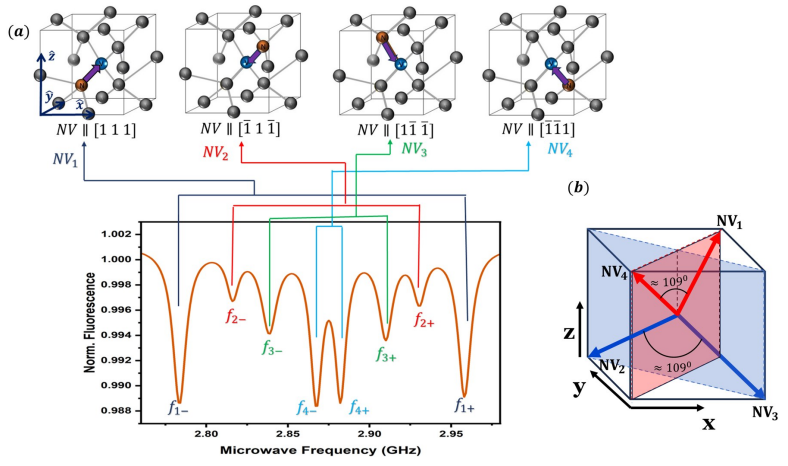


Figure 1: The four NV orientations and the ensemble ODMR spectrum

Case	Field in diamond frame	Projections	Dips
1	$\mathbf{B} = 0$	$B_1 = B_2 = B_3 = B_4 = 0$	1
2	$\mathbf{B} \parallel z$ $(B_x = B_y = 0)$	$\lvert B_i\rvert = \lvert B_z\rvert/\sqrt{3}$ for all i	2
3	$\mathbf{B} \parallel [110]$ $(B_x = B_y,\ B_z = 0)$	$B_1 = -B_4 = \frac{2B_x}{\sqrt{3}}$, $B_2 = B_3 = 0$	3
4	$B_x = B_y = B_z$ $(\mathbf{B} \parallel \mathrm{NV}_1)$	$B_1 = \sqrt{3}B_x$, $B_2 = B_3 = B_4 = -B_x/\sqrt{3}$	4

Table 1: Field configurations and the number of ODMR dips they produce.

3 Experimental arrangement and conditions

3.1 Set-up

A 532 nm laser, delivered by a single-mode fibre and fibre port, is reflected by a dichroic mirror into an air objective that focuses it onto the diamond. The red fluorescence collected by the objective passes through the dichroic mirror and a 650 nm long-pass filter and is focused by a condenser lens on a photodiode whose voltage is recorded by the DAQ and LabVIEW program. Microwaves from a signal generator, amplified by a MW amplifier, are delivered to the NV spins by a 20 μm copper wire on the diamond surface. The static field is produced by two perpendicular circular coils (150 turns each, diameter 400 mm), each with its own power supply (Fig. 2, Fig. 3).

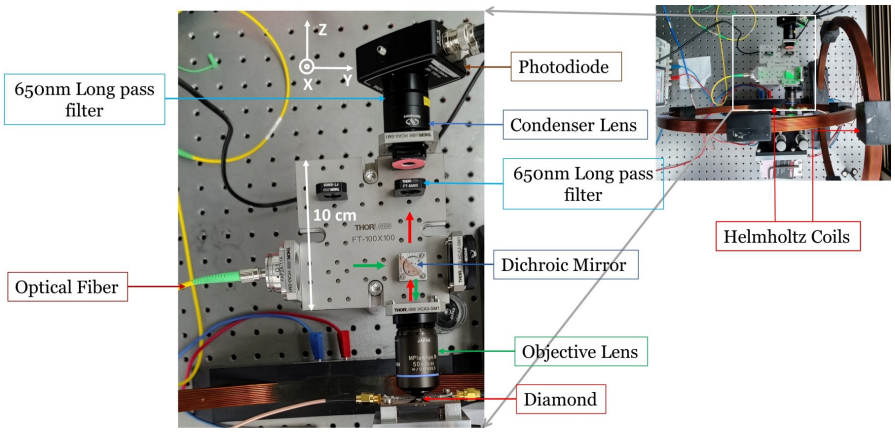


Figure 2: Diamond magnetometer: Green arrows- laser excitation, red arrows- fluorescence.

3.2 Coil geometry and reference frames

The geometry used for the field calculation is that of the sketch of our set-up (Fig. 3):

- **Coil B1** is the horizontal coil. Its axis is the lab Z axis and the diamond lies in its plane, so the axial distance is $z_Z = 0$. It produces $\mathbf{B} = (0, 0, B_Z)$.
- **Coil B2** is the vertical coil. Its axis is the lab Y axis and the diamond is $z_Y = 160$ mm from its plane. It produces $\mathbf{B} = (0, B_Y, 0)$.

Experiment confirms that with only B1 on, all four NV orientations show the same projection and exactly 2 dips appear (Case 2), whereas with only B2 on, the field points along Y axis, which is perpendicular to NV_2 and NV_3, giving a 3-dip pattern.

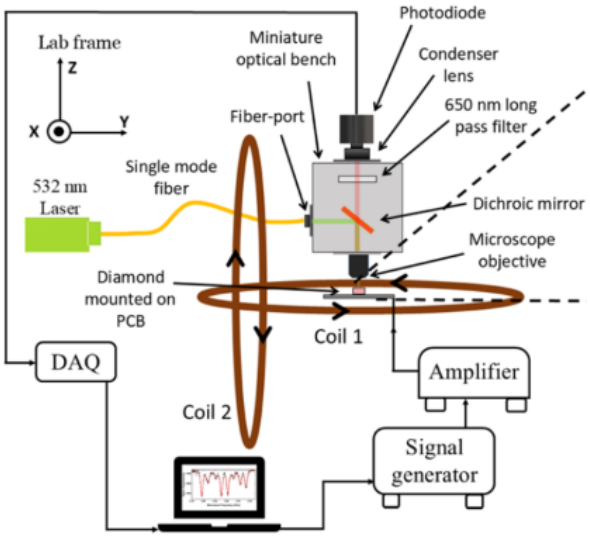


Figure 3: Experimental set-up used in the measurements. Coil 1 (B1) is horizontal (axis Z, diamond at $z = 0$), coil 2 (B2) is vertical (axis Y, diamond 160 mm from it).

The diamond is glued to the PCB with its edges rotated by 45° about Z with respect to the lab frame. The lab field $(0, B_Y, B_Z)$ therefore becomes, in the diamond frame,

$$\begin{pmatrix} B_x \\ B_y \\ B_z \end{pmatrix} = \begin{pmatrix} \cos 45^\circ & \sin 45^\circ & 0 \\ -\sin 45^\circ & \cos 45^\circ & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ B_Y \\ B_Z \end{pmatrix} = \begin{pmatrix} B_Y/\sqrt{2} \\ B_Y/\sqrt{2} \\ B_Z \end{pmatrix}. \tag{4}$$

Hence $B_x = B_y$ always holds in this set-up. Each spectrum obtained consists of 171 points (2.81–2.98 GHz, 1 MHz step) of normalised photoluminescence (PL).

3.3 Recorded data

File(s)	Series	I_1 (A)	I_2 (A)	Condition
B1_0_k	Series A	0	$k = 1 \ldots 5$	B2 only (y-axis coil)
B2_0_k	Series B	$k = 1 \ldots 5$	0	B1 only (z-axis coil)
B1_B2_4	Series C	4	4	$I_1 = I_2$
B1_B2_5_1	Series C	5	1	$I_1 \gg I_2$
B1_B2_1_5	Series C	1	5	$I_2 \gg I_1$
B1_B2_1_4_1	Series C	1.4	1	$I_1 = \sqrt{2}\, I_2$
B1_B2_1_1_4	Series C	1	1.4	$I_2 = \sqrt{2}\, I_1$ (circled in plan)
Noise	baseline	–	–	no resonance

Table 2: I_1: current in B1 (z-axis coil, $\rightarrow B_Z$); I_2: current in B2 (y-axis coil, $\rightarrow B_Y$).

4 Data analysis

4.1 Fit model

Each spectrum was fitted with a sum of Lorentzian dips on a constant baseline y_0:

$$y(f) = y_0 - \sum_k A_k \frac{(w_k/2)^2}{(f - f_k)^2 + (w_k/2)^2}, \tag{5}$$

where f_k is the resonance frequency, w_k the FWHM and A_k the contrast of dip k. The fit was carried out with nonlinear least squares; the number of dips was set from the visible structure and checked by the drop in the residual. The standard errors of f_k come from the covariance matrix and are propagated to $\Delta\nu = f_+ - f_-$ and $B_i = \Delta\nu/2\gamma$ in quadrature. All fit parameters are in Appendix A. Each spectrum has 171 points (2.81–2.98 GHz, 1 MHz step) of normalised photoluminescence (PL).

4.2 Noise level

The baseline measurement (Fig. 4) has mean 1.00012 and standard dev. $\sigma = 1.02 \times 10^{-4}$ (peak-to-peak 4.1×10^{-4}, $N = 100$). The typical dip contrast of the strong dips is $\approx 3.8\,\%$, i.e. a signal-to-noise ratio of about 374, and even the weakest dips used (contrast $\approx 0.4\,\%$) are $\gtrsim 40\sigma$. Point-to-point noise is therefore not what limits the fits: the residuals (rms 4×10^{-4}–10^{-3}) are larger than σ because practically the graphs are not perfectly Lorentzian and several dips overlap.

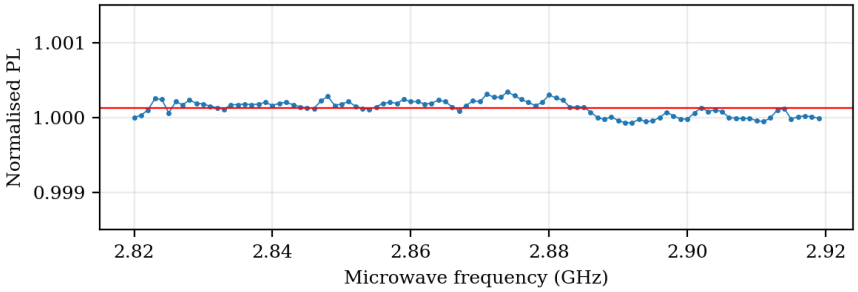


Figure 4: Baseline (no resonance) trace; the red line is the mean.

5 Part A

5.1 Q1. Components of the set-up and their function

Component	Function
532 nm laser + optical fibre	Green excitation of the NV centres ($^3A_2 \rightarrow {}^3E$); the single mode fibre and port deliver a clean collimated beam.
Dichroic mirror	Reflects green beam towards diamond and transmits red fluorescence, separating excitation and detection paths.
Microscope (air) objective	Focuses the laser on the diamond and collects part of the fluorescence emitted in all directions.
HPHT diamond on PCB	Sample with a high density of NV centres (all four orientations); the PCB carries the microwave signal.
Translation stage	Moves the PCB/diamond in three directions to focus on and position the sample while the objective stays fixed.
Copper wire (20 μm)	Delivers the microwave magnetic field to the NV spins.
Signal generator and a microwave amplifier	Generate and amplify the swept microwave signal that drives $\|0\rangle \leftrightarrow \|\pm 1\rangle$.
2 coils + 2 power supplies	Produce the static field; the two perpendicular coils with independent currents set magnitude and direction of $\mathbf{B}$.
650 nm long-pass filter	Blocks scattered green light, passing only red fluorescence.
Condenser lens	Focuses the filtered fluorescence onto the photodiode.
Photodiode	Converts the fluorescence into an analogue voltage.
DAQ + computer (LabVIEW)	Digitise the voltage, synchronise it with the frequency sweep, display and export the ODMR spectrum.
Miniature optical bench	Holds the optical elements in fixed alignment.

Table 3: Components of the diamond magnetometer.

5.2 Q2. Influence of the field projection on the number of dips

Every NV orientation i responds only to its projection $B_i = \mathbf{B} \cdot \hat{u}_i$ (Eq. 3) and gives 2 dips at $f_{i\pm} = D \pm \gamma |B_i|$, separated by $2\gamma|B_i|$. So spectrum contains one dip pair for each *distinct* value of $|B_i|$ (and single unsplit dip at D if $B_i = 0$). For $B = 0$ all projections vanish and 1 dip is seen. For $\mathbf{B} \parallel z$ all 4 projections are equal in magnitude: 2 dips. For $\mathbf{B} \parallel [110]$ two orientations (NV_2, NV_3) are perpendicular to the field and give the unsplit central dip, while NV_1 and NV_4 share the same $|B_i|$ and give the outer pair: 3 dips. If the field is along an NV axis, that orientation has 3x larger projection than the other three, which are equal to each other: 4 dips. For a general direction all 4 projections differ, 8 dips are possible (6 in our set-up, where $B_x = B_y$ forces $|B_2| = |B_3|$).

5.3 Q3. Role of the microwave field

The laser alone only polarises the spin into $|m_s = 0\rangle$ and gives a fluorescence level that does not depend on the field. Microwave field is the probe that makes the spin transition visible. Its magnetic component, applied perpendicular to the NV axis, couples $|0\rangle$ to $|\pm 1\rangle$ (matrix element of $\hat{S}_x$, $1/\sqrt{2}$), and it can do so only when its frequency matches $f_\pm = D \pm \gamma B_z$. On resonance, population is moved from the bright $|0\rangle$ state to the darker $|\pm 1\rangle$ states, which decay partly through the non-radiative singlet (intersystem-crossing) channel. Red fluorescence therefore drops slightly. Off resonance nothing is transferred and the fluorescence stays at its maximum. Sweeping the frequency while recording the PL thus gives the ODMR spectrum, and the dip positions encode the Zeeman shifts, i.e. the field projections. Dip depth and width are set by the competition between optical re-polarisation and microwave driving (power broadening), strain/electric fields, hyperfine coupling to ^{14}N and the inhomogeneity of the ensemble.

6 Q4. Current values, Magnetic field and Zeeman analysis

6.1 Coil field formula

The field on the axis of a circular loop follows from the Biot–Savart law: each element $d\mathbf{l}$ contributes $dB = \frac{\mu_0 I\, dl}{4\pi(R^2+Z^2)}$ and only the axial component $R/\sqrt{R^2+Z^2}$ survives the integration over the loop, so for N turns

$$\boxed{B(Z) = \frac{\mu_0 N I}{2}\,\frac{R^2}{(R^2+Z^2)^{3/2}}} \tag{6}$$

with $N = 150$, $R = 200$ mm (diameter 400 mm), $\mu_0 = 4\pi \times 10^{-7}$ T m/A, I in A and Z the axial distance from the plane of the coil. The field is calculated per ampere:

$$k_Z = \frac{\mu_0 N R^2}{2\,(R^2+z_Z^2)^{3/2}}\bigg|_{z_Z=0} = \frac{\mu_0 N}{2R} = \frac{(4\pi\times10^{-7})(150)}{2(0.200)} = \mathbf{0.4712\ mT/A},$$

$$k_Y = \frac{\mu_0 N R^2}{2\,(R^2+z_Y^2)^{3/2}}\bigg|_{z_Y=0.160\ \mathrm{m}} = \frac{(4\pi\times10^{-7})(150)(0.200)^2}{2\left[(0.200)^2+(0.160)^2\right]^{3/2}}$$
$$= 2.2438\times10^{-4}\ \mathrm{T/A} = \mathbf{0.2244\ mT/A}.$$

Hence, with $B_Z = k_Z I_Z$ and $B_Y = k_Y I_Y$ in mT. (Units used throughout: current in A, B in mT, $\gamma = 28$ MHz/mT, $k_Z = 0.4712 \times 10^{-3}$ T/A.)

6.2 Field vector in the diamond frame and NV projections

For every current pair the following steps are applied: $B_Z = k_Z I_Z$, $B_Y = k_Y I_Y$; Eq. 4 gives $B_x = B_y = B_Y/\sqrt{2}$ and $B_z = B_Z$; Eq. 3 gives $B_1, \ldots, B_4$; the predicted splitting of orientation i is $2\gamma|B_i| = 56|B_i|$ MHz. These steps are written out below, together with the measured quantities $\Delta\nu = f_+ - f_-$ and $B_i = \Delta\nu/2\gamma$.

6.3 Case 1: one dip (B = 0)

For $I_Y = I_Z = 0$: $B_Z = B_Y = 0$, hence $B_x = B_y = B_z = 0$ and $B_1 = B_2 = B_3 = B_4 = 0$. A single dip at $D = 2.870$ GHz is expected, possibly with a small splitting from strain and electric fields. This case is illustrated by the weakest runs of Series A below ($I_Y = 1, 2$ A, $B_Y \leq 0.45$ mT), where only the central doublet is visible (Fig. 6). Its two components lie 5.7–7.3 MHz apart almost independently of the current; this is the strain/electric splitting $2E$ of the $|\pm 1\rangle$ levels at zero field ($E \approx 3.2$ MHz).

6.4 Case 2: two dips (field along z) – Series B

Only coil B1 carries current, $I_Y = 0$, so $B_x = B_y = 0$ and all four projections equal $B_Z/\sqrt{3}$. $I_Z = 1, \ldots, 5$ A (five values, splitting increasing; Fig. 5).

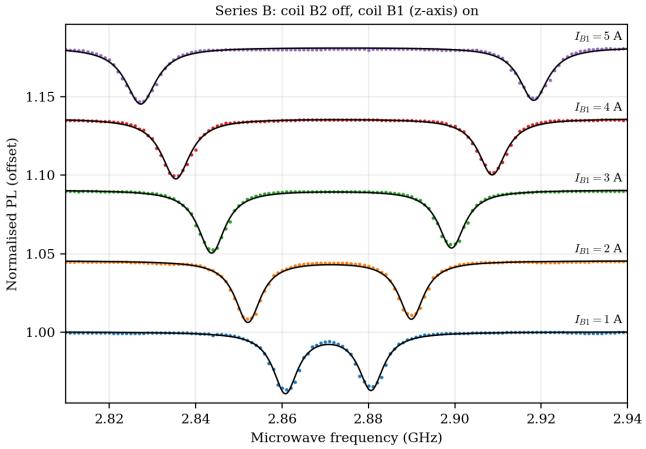


Figure 5: Case 2: 2 dips, whose splitting grows with I_Z. black curve: Lorentzian fits.

$I_Z = 1$ **A,** $I_Y = 0$

$$
\begin{aligned}
B_Z &= k_Z I_Z = 0.4712 \times 1 = 0.4712 \text{ mT} \\
B_Y &= k_Y I_Y = 0.2244 \times 0 = 0.0000 \text{ mT} \\
B_x = B_y &= B_Y/\sqrt{2} = 0.0000/1.4142 = 0.0000 \text{ mT} \\
B_1 &= (B_x + B_y + B_z)/\sqrt{3} = (0.0000 + 0.0000 + 0.4712)/1.7321 = 0.2721 \text{ mT} \\
B_2 &= (-B_x + B_y - B_z)/\sqrt{3} = (-0.0000 + 0.0000 - 0.4712)/1.7321 = -0.2721 \text{ mT} \\
B_3 &= (B_x - B_y - B_z)/\sqrt{3} = (0.0000 - 0.0000 - 0.4712)/1.7321 = -0.2721 \text{ mT} \\
B_4 &= (-B_x - B_y + B_z)/\sqrt{3} = (-0.0000 - 0.0000 + 0.4712)/1.7321 = 0.2721 \text{ mT} \\
\Delta\nu_i^{\text{th}} &= 56\,|B_i| = (15.2,\ 15.2,\ 15.2,\ 15.2) \text{ MHz}
\end{aligned}
$$

Experiment:

$$
\begin{aligned}
f_- &= 2.8608 \text{ GHz}, \quad f_+ = 2.8806 \text{ GHz} \\
\Delta\nu &= (f_+ - f_-) \times 10^3 = 19.82 \pm 0.07 \text{ MHz} \\
B_i &= \frac{\Delta\nu}{2\gamma} = \frac{19.82}{56} = 0.354 \pm 0.001 \text{ mT}
\end{aligned}
$$

$I_Z = 2$ **A,** $I_Y = 0$

$$
\begin{aligned}
B_Z &= k_Z I_Z = 0.4712 \times 2 = 0.9425 \text{ mT} \\
B_Y &= k_Y I_Y = 0.2244 \times 0 = 0.0000 \text{ mT} \\
B_x = B_y &= B_Y/\sqrt{2} = 0.0000/1.4142 = 0.0000 \text{ mT} \\
B_1 &= (B_x + B_y + B_z)/\sqrt{3} = (0.0000 + 0.0000 + 0.9425)/1.7321 = 0.5441 \text{ mT} \\
B_2 &= (-B_x + B_y - B_z)/\sqrt{3} = (-0.0000 + 0.0000 - 0.9425)/1.7321 = -0.5441 \text{ mT} \\
B_3 &= (B_x - B_y - B_z)/\sqrt{3} = (0.0000 - 0.0000 - 0.9425)/1.7321 = -0.5441 \text{ mT} \\
B_4 &= (-B_x - B_y + B_z)/\sqrt{3} = (-0.0000 - 0.0000 + 0.9425)/1.7321 = 0.5441 \text{ mT} \\
\Delta\nu_i^{\text{th}} &= 56\,|B_i| = (30.5,\ 30.5,\ 30.5,\ 30.5) \text{ MHz}
\end{aligned}
$$

Experiment:

$$
\begin{aligned}
f_- &= 2.8521 \text{ GHz}, \quad f_+ = 2.8900 \text{ GHz} \\
\Delta\nu &= (f_+ - f_-) \times 10^3 = 37.83 \pm 0.07 \text{ MHz} \\
B_i &= \frac{\Delta\nu}{2\gamma} = \frac{37.83}{56} = 0.676 \pm 0.001 \text{ mT}
\end{aligned}
$$

$I_Z = 3$ **A,** $I_Y = 0$

$$
\begin{aligned}
B_Z &= k_Z I_Z = 0.4712 \times 3 = 1.4137 \text{ mT} \\
B_Y &= k_Y I_Y = 0.2244 \times 0 = 0.0000 \text{ mT} \\
B_x = B_y &= B_Y/\sqrt{2} = 0.0000/1.4142 = 0.0000 \text{ mT} \\
B_1 &= (B_x + B_y + B_z)/\sqrt{3} = (0.0000 + 0.0000 + 1.4137)/1.7321 = 0.8162 \text{ mT} \\
B_2 &= (-B_x + B_y - B_z)/\sqrt{3} = (-0.0000 + 0.0000 - 1.4137)/1.7321 = -0.8162 \text{ mT} \\
B_3 &= (B_x - B_y - B_z)/\sqrt{3} = (0.0000 - 0.0000 - 1.4137)/1.7321 = -0.8162 \text{ mT} \\
B_4 &= (-B_x - B_y + B_z)/\sqrt{3} = (-0.0000 - 0.0000 + 1.4137)/1.7321 = 0.8162 \text{ mT} \\
\Delta\nu_i^{\text{th}} &= 56\,|B_i| = (45.7,\ 45.7,\ 45.7,\ 45.7) \text{ MHz}
\end{aligned}
$$

Experiment:

$$f_- = 2.8437 \text{ GHz}, \quad f_+ = 2.8993 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 55.52 \pm 0.07 \text{ MHz}$$
$$B_i = \frac{\Delta\nu}{2\gamma} = \frac{55.52}{56} = 0.991 \pm 0.001 \text{ mT}$$

$I_Z = 4$ **A,** $I_Y = 0$

$$B_Z = k_Z I_Z = 0.4712 \times 4 = 1.8850 \text{ mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 0 = 0.0000 \text{ mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 0.0000/1.4142 = 0.0000 \text{ mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.0000 + 0.0000 + 1.8850)/1.7321 = 1.0883 \text{ mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.0000 + 0.0000 - 1.8850)/1.7321 = -1.0883 \text{ mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.0000 - 0.0000 - 1.8850)/1.7321 = -1.0883 \text{ mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.0000 - 0.0000 + 1.8850)/1.7321 = 1.0883 \text{ mT}$$
$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (60.9,\ 60.9,\ 60.9,\ 60.9) \text{ MHz}$$

Experiment:

$$f_- = 2.8355 \text{ GHz}, \quad f_+ = 2.9087 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 73.20 \pm 0.07 \text{ MHz}$$
$$B_i = \frac{\Delta\nu}{2\gamma} = \frac{73.20}{56} = 1.307 \pm 0.001 \text{ mT}$$

$I_Z = 5$ **A,** $I_Y = 0$

$$B_Z = k_Z I_Z = 0.4712 \times 5 = 2.3562 \text{ mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 0 = 0.0000 \text{ mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 0.0000/1.4142 = 0.0000 \text{ mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.0000 + 0.0000 + 2.3562)/1.7321 = 1.3603 \text{ mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.0000 + 0.0000 - 2.3562)/1.7321 = -1.3603 \text{ mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.0000 - 0.0000 - 2.3562)/1.7321 = -1.3603 \text{ mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.0000 - 0.0000 + 2.3562)/1.7321 = 1.3603 \text{ mT}$$
$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (76.2,\ 76.2,\ 76.2,\ 76.2) \text{ MHz}$$

Experiment:

$$f_- = 2.8273 \text{ GHz}, \quad f_+ = 2.9183 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 90.92 \pm 0.08 \text{ MHz}$$
$$B_i = \frac{\Delta\nu}{2\gamma} = \frac{90.92}{56} = 1.624 \pm 0.001 \text{ mT}$$

Peaks	I_Y **(A)**	I_Z **(A)**	(B_x, B_y, B_z) **(mT)**	B_1 **(mT)**	B_2 **(mT)**	B_3 **(mT)**	B_4 **(mT)**
2	0	1	(0.000, 0.000, 0.471)	0.354	0.354	0.354	0.354
2	0	2	(0.000, 0.000, 0.942)	0.676	0.676	0.676	0.676
2	0	3	(0.000, 0.000, 1.414)	0.991	0.991	0.991	0.991
2	0	4	(0.000, 0.000, 1.885)	1.307	1.307	1.307	1.307
2	0	5	(0.000, 0.000, 2.356)	1.624	1.624	1.624	1.624

Table 4: Series B (Case 2, two dips): B2 off, B1 (z-axis coil) on.

I_Y (A)	I_Z (A)	$\lvert B_1\rvert^{\text{th}}$	$\lvert B_2\rvert^{\text{th}}$	$\lvert B_3\rvert^{\text{th}}$	$\lvert B_4\rvert^{\text{th}}$	dips th./obs.
0	1	0.272	0.272	0.272	0.272	2 / 2
0	2	0.544	0.544	0.544	0.544	2 / 2
0	3	0.816	0.816	0.816	0.816	2 / 2
0	4	1.088	1.088	1.088	1.088	2 / 2
0	5	1.360	1.360	1.360	1.360	2 / 2

Table 5: Theoretical Series B (all projections in mT).

I_Y (A)	I_Z (A)	$\Delta\nu^{\text{th}}$ (MHz)	$\Delta\nu^{\text{exp}}$ (MHz)	exp/th
0	1	15.2	19.8 ± 0.1	1.30
0	2	30.5	37.8 ± 0.1	1.24
0	3	45.7	55.5 ± 0.1	1.21
0	4	60.9	73.2 ± 0.1	1.20
0	5	76.2	90.9 ± 0.1	1.19

Table 6: Series B: predicted vs. measured splitting of the observed dip pair.

6.5 Case 3: three dips (field along [110]) – Series A

Only coil B2 carries current, $I_Z = 0$, so $B_z = 0$, $B_x = B_y = B_Y/\sqrt{2}$, $B_2 = B_3 = 0$ and $B_1 = -B_4 = \sqrt{2/3}\,B_Y$. The three-dip pattern (outer pair from NV_1/NV_4, central dip from NV_2/NV_3) is resolved for $I_Y = 3, 4, 5$ A (three distinct currents with increasing splitting); at 1 and 2 A the outer dips are still inside the central dip (Fig. 6). The central dip is itself split into a doublet (strain/electric fields), which is not a Zeeman effect.

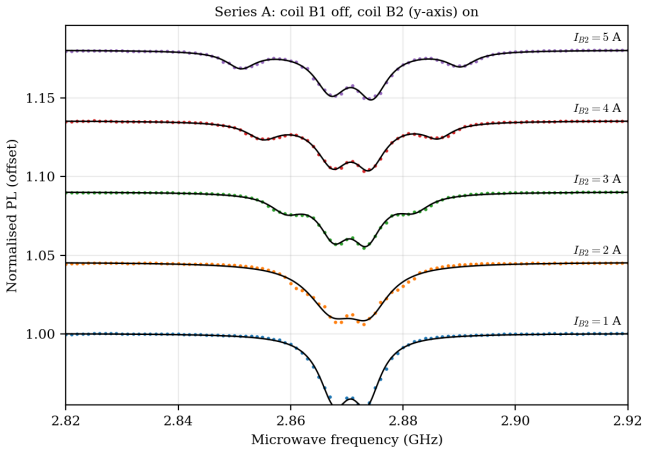


Figure 6: B2 only:(NV_1, NV_4) dips separate with current, central dip (NV_2, NV_3) stays at D

$I_Z = 0$**,** $I_Y = 1$ **A**

$$B_Z = k_Z I_Z = 0.4712 \times 0 = 0.0000 \text{ mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 1 = 0.2244 \text{ mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 0.2244/1.4142 = 0.1587 \text{ mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.1587 + 0.1587 + 0.0000)/1.7321 = 0.1832 \text{ mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.1587 + 0.1587 - 0.0000)/1.7321 = 0.0000 \text{ mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.1587 - 0.1587 - 0.0000)/1.7321 = -0.0000 \text{ mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.1587 - 0.1587 + 0.0000)/1.7321 = -0.1832 \text{ mT}$$
$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (10.3,\ 0.0,\ 0.0,\ 10.3) \text{ MHz}$$

Experiment:

$$\Delta\nu_{\text{centre}} = (2.8735 - 2.8678) \times 10^3 = 5.69 \text{ MHz} \quad \text{(central doublet, not Zeeman)}$$

outer (NV_1, NV_4) dips not resolved from the central dip

$I_Z = 0$, $I_Y = 2$ A

$$B_Z = k_Z I_Z = 0.4712 \times 0 = 0.0000 \text{ mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 2 = 0.4488 \text{ mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 0.4488/1.4142 = 0.3173 \text{ mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.3173 + 0.3173 + 0.0000)/1.7321 = 0.3664 \text{ mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.3173 + 0.3173 - 0.0000)/1.7321 = 0.0000 \text{ mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.3173 - 0.3173 - 0.0000)/1.7321 = -0.0000 \text{ mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.3173 - 0.3173 + 0.0000)/1.7321 = -0.3664 \text{ mT}$$
$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (20.5,\ 0.0,\ 0.0,\ 20.5) \text{ MHz}$$

Experiment:

$$\Delta\nu_{\text{centre}} = (2.8738 - 2.8673) \times 10^3 = 6.55 \text{ MHz} \quad \text{(central doublet, not Zeeman)}$$

outer (NV_1, NV_4) dips not resolved from the central dip

$I_Z = 0$, $I_Y = 3$ A

$$B_Z = k_Z I_Z = 0.4712 \times 0 = 0.0000 \text{ mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 3 = 0.6731 \text{ mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 0.6731/1.4142 = 0.4760 \text{ mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.4760 + 0.4760 + 0.0000)/1.7321 = 0.5496 \text{ mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.4760 + 0.4760 - 0.0000)/1.7321 = 0.0000 \text{ mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.4760 - 0.4760 - 0.0000)/1.7321 = -0.0000 \text{ mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.4760 - 0.4760 + 0.0000)/1.7321 = -0.5496 \text{ mT}$$
$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (30.8,\ 0.0,\ 0.0,\ 30.8) \text{ MHz}$$

Experiment:

$$f_- = 2.8592 \text{ GHz}, \quad f_+ = 2.8820 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 22.75 \pm 0.18 \text{ MHz}$$
$$B_1 = B_4 = \frac{\Delta\nu}{2\gamma} = \frac{22.75}{56} = 0.406 \pm 0.003 \text{ mT}$$
$$\Delta\nu_{\text{centre}} = (2.8735 - 2.8677) \times 10^3 = 5.81 \text{ MHz} \quad \text{(central doublet, not Zeeman)}$$

$I_Z = 0$, $I_Y = 4$ A

$$B_Z = k_Z I_Z = 0.4712 \times 0 = 0.0000 \text{ mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 4 = 0.8975 \text{ mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 0.8975/1.4142 = 0.6346 \text{ mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.6346 + 0.6346 + 0.0000)/1.7321 = 0.7328 \text{ mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.6346 + 0.6346 - 0.0000)/1.7321 = 0.0000 \text{ mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.6346 - 0.6346 - 0.0000)/1.7321 = -0.0000 \text{ mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.6346 - 0.6346 + 0.0000)/1.7321 = -0.7328 \text{ mT}$$

$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (41.0,\ 0.0,\ 0.0,\ 41.0)\ \text{MHz}$$

Experiment:

$$f_- = 2.8551\ \text{GHz}, \quad f_+ = 2.8862\ \text{GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 31.11 \pm 0.13\ \text{MHz}$$
$$B_1 = B_4 = \frac{\Delta\nu}{2\gamma} = \frac{31.11}{56} = 0.556 \pm 0.002\ \text{mT}$$
$$\Delta\nu_{\text{centre}} = (2.8741 - 2.8675) \times 10^3 = 6.61\ \text{MHz} \quad \text{(central doublet, not Zeeman)}$$

$I_Z = 0$, $I_Y = 5$ A

$$B_Z = k_Z I_Z = 0.4712 \times 0 = 0.0000\ \text{mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 5 = 1.1219\ \text{mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 1.1219/1.4142 = 0.7933\ \text{mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.7933 + 0.7933 + 0.0000)/1.7321 = 0.9160\ \text{mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.7933 + 0.7933 - 0.0000)/1.7321 = 0.0000\ \text{mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.7933 - 0.7933 - 0.0000)/1.7321 = -0.0000\ \text{mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.7933 - 0.7933 + 0.0000)/1.7321 = -0.9160\ \text{mT}$$
$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (51.3,\ 0.0,\ 0.0,\ 51.3)\ \text{MHz}$$

Experiment:

$$f_- = 2.8512\ \text{GHz}, \quad f_+ = 2.8902\ \text{GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 39.08 \pm 0.15\ \text{MHz}$$
$$B_1 = B_4 = \frac{\Delta\nu}{2\gamma} = \frac{39.08}{56} = 0.698 \pm 0.003\ \text{mT}$$
$$\Delta\nu_{\text{centre}} = (2.8745 - 2.8672) \times 10^3 = 7.34\ \text{MHz} \quad \text{(central doublet, not Zeeman)}$$

Peaks	I_Y (A)	I_Z (A)	(B_x, B_y, B_z) (mT)	B_1 (mT)	B_2 (mT)	B_3 (mT)	B_4 (mT)
1‡	1	0	(0.159, 0.159, 0.000)	n.r.	n.r.	n.r.	n.r.
1‡	2	0	(0.317, 0.317, 0.000)	n.r.	n.r.	n.r.	n.r.
3	3	0	(0.476, 0.476, 0.000)	0.406	≈ 0	≈ 0	0.406
3	4	0	(0.635, 0.635, 0.000)	0.556	≈ 0	≈ 0	0.556
3	5	0	(0.793, 0.793, 0.000)	0.698	≈ 0	≈ 0	0.698

Table 7: Case 3: B1 off, B2 on. I_Y = current in B2; I_Z = current in B1. $B_i = \Delta\nu_i/2\gamma$ from the fits. n.r. = not resolved; ‡ only central strain doublet visible.

I_Y (A)	I_Z (A)	$\lvert B_1\rvert^{\text{th}}$	$\lvert B_2\rvert^{\text{th}}$	$\lvert B_3\rvert^{\text{th}}$	$\lvert B_4\rvert^{\text{th}}$	dips th./obs.
1	0	0.183	0.000	0.000	0.183	3 / 1‡
2	0	0.366	0.000	0.000	0.366	3 / 1‡
3	0	0.550	0.000	0.000	0.550	3 / 3
4	0	0.733	0.000	0.000	0.733	3 / 3
5	0	0.916	0.000	0.000	0.916	3 / 3

Table 8: Theory (corrected coil formula), Series A (all projections in mT).

I_Y (A)	I_Z (A)	$\Delta\nu^{\text{th}}$ (MHz)	$\Delta\nu^{\text{exp}}$ (MHz)	exp/th
3	0	30.8	22.8 ± 0.2	0.74
4	0	41.0	31.1 ± 0.1	0.76
5	0	51.3	39.1 ± 0.1	0.76

Table 9: Series A: predicted vs. measured splitting of the outer (NV_1/NV_4) dip pair.

6.6 Case 4 and mixed field configurations – Series C

Both coils are on, 5 five current pairs of the measurement plan were recorded (Table 2).

Condition for Case 4. Case 4 requires $B_x = B_y = B_z$, i.e. $B_Y/\sqrt{2} = B_Z$:

$$\text{Condition } B_x = B_y = B_z: \quad \frac{B_Y}{\sqrt{2}} = B_Z \;\Rightarrow\; k_Y I_Y = \sqrt{2}\, k_Z I_Z$$

$$\Rightarrow \quad \frac{I_Y}{I_Z} = \frac{\sqrt{2}\, k_Z}{k_Y} = \frac{1.4142 \times 0.4712}{0.2244} = \mathbf{2.97}$$

$$(\text{naive } k_Y = k_Z: I_Y/I_Z = \sqrt{2} = 1.41)$$

So $I_Y \approx 2.97\, I_Z$ is needed (not $\sqrt{2}\, I_Z$): the y-axis coil is 160 mm from the diamond and is only $k_Y/k_Z = 0.476$ as effective as the z-axis coil. Consequently the pair $(I_1, I_2) = (1, 1.4)$ A circled in the plan (and its mirror $(1.4, 1)$ A) does *not* satisfy the condition: the field is dominated by B_Z and only two (broadened) dips are seen. The two series C runs with a substantial B_Y give the richer spectra (Fig. 7).

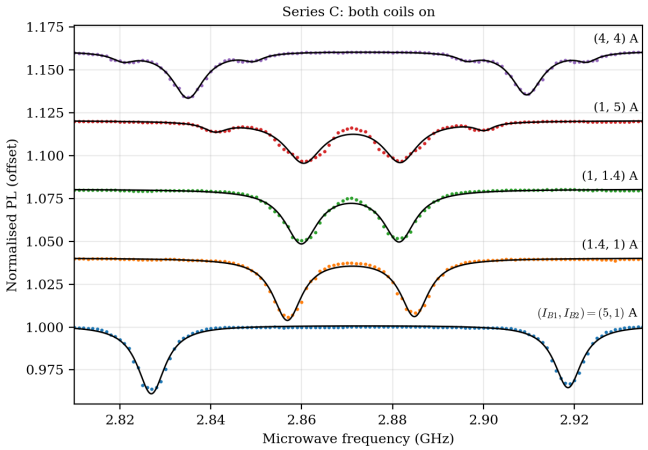


Figure 7: Series C: both coils on, currents (I_{B1}, I_{B2}) indicated. Black dots: Lorentzian fits.

$I_Z = 4$ **A,** $I_Y = 4$ **A**

$$B_Z = k_Z I_Z = 0.4712 \times 4 = 1.8850 \text{ mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 4 = 0.8975 \text{ mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 0.8975/1.4142 = 0.6346 \text{ mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.6346 + 0.6346 + 1.8850)/1.7321 = 1.8211 \text{ mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.6346 + 0.6346 - 1.8850)/1.7321 = -1.0883 \text{ mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.6346 - 0.6346 - 1.8850)/1.7321 = -1.0883 \text{ mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.6346 - 0.6346 + 1.8850)/1.7321 = 0.3555 \text{ mT}$$
$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (102.0,\ 60.9,\ 60.9,\ 19.9) \text{ MHz}$$

Experiment:

$$f_- = 2.8207 \text{ GHz}, \quad f_+ = 2.9228 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 102.05 \pm 0.30 \text{ MHz}$$
$$B_1 = \frac{\Delta\nu}{2\gamma} = \frac{102.05}{56} = 1.822 \pm 0.005 \text{ mT}$$
$$f_- = 2.8350 \text{ GHz}, \quad f_+ = 2.9095 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 74.53 \pm 0.06 \text{ MHz}$$

$$B_2 = B_3 = \frac{\Delta\nu}{2\gamma} = \frac{74.53}{56} = 1.331 \pm 0.001 \text{ mT}$$
$$f_- = 2.8494 \text{ GHz}, \quad f_+ = 2.8964 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 46.99 \pm 0.34 \text{ MHz}$$
$$B_4 = \frac{\Delta\nu}{2\gamma} = \frac{46.99}{56} = 0.839 \pm 0.006 \text{ mT}$$

$I_Z = 5$ **A,** $I_Y = 1$ **A**

$$B_Z = k_Z I_Z = 0.4712 \times 5 = 2.3562 \text{ mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 1 = 0.2244 \text{ mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 0.2244/1.4142 = 0.1587 \text{ mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.1587 + 0.1587 + 2.3562)/1.7321 = 1.5436 \text{ mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.1587 + 0.1587 - 2.3562)/1.7321 = -1.3603 \text{ mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.1587 - 0.1587 - 2.3562)/1.7321 = -1.3603 \text{ mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.1587 - 0.1587 + 2.3562)/1.7321 = 1.1771 \text{ mT}$$
$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (86.4,\ 76.2,\ 76.2,\ 65.9) \text{ MHz}$$

Experiment:

$$f_- = 2.8269 \text{ GHz}, \quad f_+ = 2.9186 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 91.69 \pm 0.07 \text{ MHz}$$
$$B_i = \frac{\Delta\nu}{2\gamma} = \frac{91.69}{56} = 1.637 \pm 0.001 \text{ mT}$$

$I_Z = 1$ **A,** $I_Y = 5$ **A**

$$B_Z = k_Z I_Z = 0.4712 \times 1 = 0.4712 \text{ mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 5 = 1.1219 \text{ mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 1.1219/1.4142 = 0.7933 \text{ mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.7933 + 0.7933 + 0.4712)/1.7321 = 1.1881 \text{ mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.7933 + 0.7933 - 0.4712)/1.7321 = -0.2721 \text{ mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.7933 - 0.7933 - 0.4712)/1.7321 = -0.2721 \text{ mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.7933 - 0.7933 + 0.4712)/1.7321 = -0.6439 \text{ mT}$$
$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (66.5,\ 15.2,\ 15.2,\ 36.1) \text{ MHz}$$

Experiment:

$$f_- = 2.8410 \text{ GHz}, \quad f_+ = 2.9002 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 59.18 \pm 0.58 \text{ MHz}$$
$$B_1 = \frac{\Delta\nu}{2\gamma} = \frac{59.18}{56} = 1.057 \pm 0.010 \text{ mT}$$
$$f_- = 2.8604 \text{ GHz}, \quad f_+ = 2.8817 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 21.30 \pm 0.16 \text{ MHz}$$
$$B_2 \approx B_3 \approx B_4 = \frac{\Delta\nu}{2\gamma} = \frac{21.30}{56} = 0.380 \pm 0.003 \text{ mT}$$

$I_Z = 1.4$ **A,** $I_Y = 1$ **A**

$$B_Z = k_Z I_Z = 0.4712 \times 1.4 = 0.6597 \text{ mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 1 = 0.2244 \text{ mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 0.2244/1.4142 = 0.1587 \text{ mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.1587 + 0.1587 + 0.6597)/1.7321 = 0.5641 \text{ mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.1587 + 0.1587 - 0.6597)/1.7321 = -0.3809 \text{ mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.1587 - 0.1587 - 0.6597)/1.7321 = -0.3809 \text{ mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.1587 - 0.1587 + 0.6597)/1.7321 = 0.1977 \text{ mT}$$
$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (31.6,\ 21.3,\ 21.3,\ 11.1) \text{ MHz}$$

Experiment:

$$f_- = 2.8569 \text{ GHz}, \quad f_+ = 2.8848 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 27.95 \pm 0.08 \text{ MHz}$$
$$B_i = \frac{\Delta\nu}{2\gamma} = \frac{27.95}{56} = 0.499 \pm 0.001 \text{ mT}$$

$I_Z = 1$ **A,** $I_Y = 1.4$ **A**

$$B_Z = k_Z I_Z = 0.4712 \times 1 = 0.4712 \text{ mT}$$
$$B_Y = k_Y I_Y = 0.2244 \times 1.4 = 0.3141 \text{ mT}$$
$$B_x = B_y = B_Y/\sqrt{2} = 0.3141/1.4142 = 0.2221 \text{ mT}$$
$$B_1 = (B_x + B_y + B_z)/\sqrt{3} = (0.2221 + 0.2221 + 0.4712)/1.7321 = 0.5286 \text{ mT}$$
$$B_2 = (-B_x + B_y - B_z)/\sqrt{3} = (-0.2221 + 0.2221 - 0.4712)/1.7321 = -0.2721 \text{ mT}$$
$$B_3 = (B_x - B_y - B_z)/\sqrt{3} = (0.2221 - 0.2221 - 0.4712)/1.7321 = -0.2721 \text{ mT}$$
$$B_4 = (-B_x - B_y + B_z)/\sqrt{3} = (-0.2221 - 0.2221 + 0.4712)/1.7321 = 0.0156 \text{ mT}$$
$$\Delta\nu_i^{\text{th}} = 56\,|B_i| = (29.6,\ 15.2,\ 15.2,\ 0.9) \text{ MHz}$$

Experiment:

$$f_- = 2.8599 \text{ GHz}, \quad f_+ = 2.8815 \text{ GHz}$$
$$\Delta\nu = (f_+ - f_-) \times 10^3 = 21.54 \pm 0.09 \text{ MHz}$$
$$B_i = \frac{\Delta\nu}{2\gamma} = \frac{21.54}{56} = 0.385 \pm 0.002 \text{ mT}$$

Peaks	I_Y **(A)**	I_Z **(A)**	(B_x, B_y, B_z) **(mT)**	B_1 **(mT)**	B_2 **(mT)**	B_3 **(mT)**	B_4 **(mT)**
6	4	4	(0.635, 0.635, 1.885)	1.822	1.331	1.331	0.839
2†	1	5	(0.159, 0.159, 2.356)	1.637	1.637	1.637	1.637
4	5	1	(0.793, 0.793, 0.471)	1.057	0.380	0.380	0.380
2†	1	1.4	(0.159, 0.159, 0.660)	0.499	0.499	0.499	0.499
2†	1.4	1	(0.222, 0.222, 0.471)	0.385	0.385	0.385	0.385

I_Y **(A)**	I_Z **(A)**	$\lvert B_1\rvert^{\text{th}}$	$\lvert B_2\rvert^{\text{th}}$	$\lvert B_3\rvert^{\text{th}}$	$\lvert B_4\rvert^{\text{th}}$	**dips th./obs.**
4	4	1.821	1.088	1.088	0.355	6 / 6
1	5	1.544	1.360	1.360	1.177	6 / 2†
5	1	1.188	0.272	0.272	0.644	6 / 4
1	1.4	0.564	0.381	0.381	0.198	6 / 2†
1.4	1	0.529	0.272	0.272	0.016	6 / 2†

Table 10: Series C: both coils on. † the four NV projections are not resolved; the tabulated value is the half-splitting of the merged dip pair (all projections in mT).

Case-4 spectrum. The spectrum with $(I_Z, I_Y) = (1, 5)$ A is used. It shows four dips (Fig. 8): an outer pair (NV_1) and an inner pair in which the three equal projections (NV_2, NV_3, NV_4) are merged. We evaluated $\mathbf{B} = (0.793, 0.793, 0.471)$ mT, i.e. $B_x/B_z = 1.68$, which is not Case 4. The measured pattern is explained by the experimental coil constants with k_Z^{exp} and k_Y^{exp} the field is almost exactly along NV_1:

$$
\begin{aligned}
B_z &= k_Z^{\text{exp}} I_Z = 0.568 \times 1 = 0.568 \text{ mT} \\
B_Y &= k_Y^{\text{exp}} I_Y = 0.170 \times 5 = 0.849 \text{ mT} \\
B_x = B_y &= B_Y/\sqrt{2} = 0.600 \text{ mT} \quad \Rightarrow \quad B_x/B_z = 1.06 \approx 1. \\
B_1 &= \frac{B_x + B_y + B_z}{\sqrt{3}} = \frac{0.600 + 0.600 + 0.568}{1.7321} = 1.021 \text{ mT} \\
&\Rightarrow \Delta\nu_1 = 57.2 \text{ MHz} \quad \text{(measured 59.2 MHz)} \\
|B_2| = |B_3| &= \frac{B_z}{\sqrt{3}} = 0.328 \text{ mT} \;\Rightarrow\; \Delta\nu = 18.4 \text{ MHz} \\
|B_4| &= \frac{|B_z - 2B_x|}{\sqrt{3}} = 0.365 \text{ mT} \;\Rightarrow\; \Delta\nu = 20.5 \text{ MHz} \\
&\text{(these three merge; measured merged pair: 21.3 MHz)}
\end{aligned}
$$

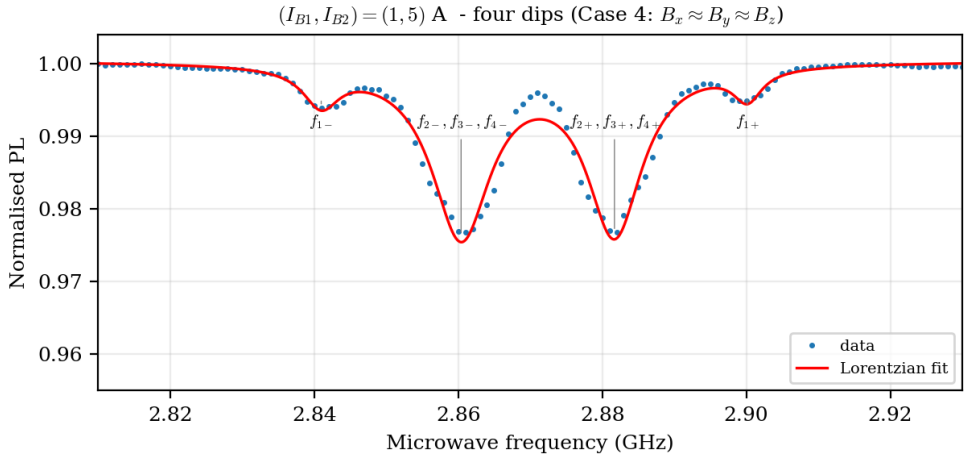


Figure 8: Case-4 spectrum, $(I_Z, I_Y) = (1, 5)$ A.

Spectrum with $(4, 4)$ A. The spectrum with both currents 4 A shows *six* dips (Fig. 9): three distinct projections, as predicted for a general field with $B_x = B_y$. The three splittings are 102.1, 74.5 and 47.0 MHz. In our geometry $|B_2| = |B_3| = |B_z|/\sqrt{3}$ and $B_1 + B_4 = 2B_2$ (signed), so the middle splitting must be the mean of the other two: $(102.1 + 47.0)/2 = 74.6$ MHz, in agreement with the measured 74.5 MHz. This confirms the assignment (outer pair NV_1, middle NV_2/NV_3, inner NV_4) and allows a complete reconstruction of the field vector from the data alone:

$$
\begin{aligned}
B_2 = B_3 &= \frac{B_1 + B_4}{2} = \frac{1.822 + 0.839}{2} = 1.331 \text{ mT} \quad \text{(measured 1.331)} \\
B_z &= \sqrt{3}\, B_2 = 1.7321 \times 1.331 = 2.305 \text{ mT} \\
B_x = B_y &= \frac{\sqrt{3}\, B_1 - B_z}{2} = \frac{1.7321 \times 1.822 - 2.305}{2} = 0.426 \text{ mT} \\
B_Y &= \sqrt{2}\, B_x = 0.602 \text{ mT} \\
B_4^{\text{pred}} &= \frac{|B_z - 2B_x|}{\sqrt{3}} = \frac{|2.305 - 0.851|}{1.7321} = 0.839 \text{ mT} \quad \text{(measured 0.839)} \\
k_Z^{\text{rec}} &= \frac{B_z}{I_Z} = \frac{2.305}{4} = 0.576 \text{ mT/A} \\
k_Y^{\text{rec}} &= \frac{B_Y}{I_Y} = \frac{0.602}{4} = 0.150 \text{ mT/A}
\end{aligned}
$$

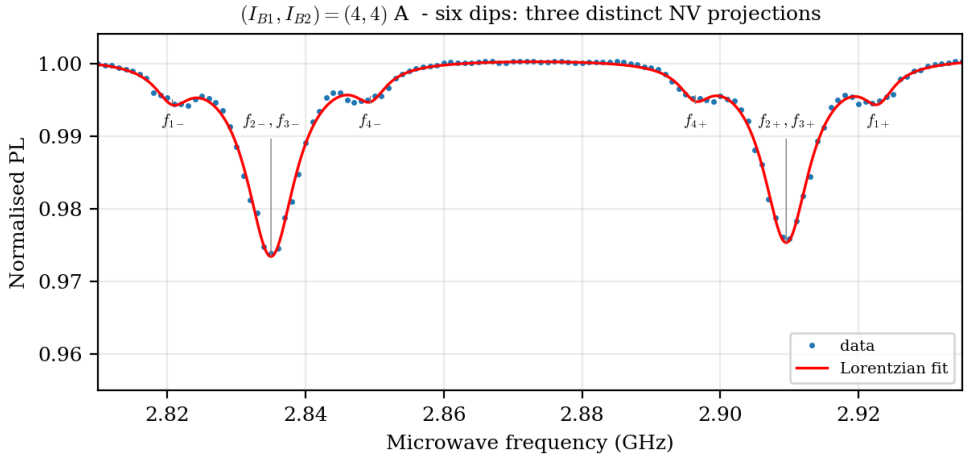


Figure 9: Spectrum for $I_Z = I_Y = 4$ A with the six dips labelled by NV orientation.

6.7 Is there another way to get four dips?

Yes. Four dips also occur when the two NV pairs have different projections but the same projection within each pair. Pair NV_1/NV_4 has equal $|B_i|$ if $B_z = 0$ (or $B_x = -B_y$); pair NV_2/NV_3 has equal $|B_i|$ if $B_z = 0$ (or $B_x = B_y$). For $B_z = 0$ both conditions hold.

$$|B_1| = |B_4| = \frac{|B_x + B_y|}{\sqrt{3}}, \qquad |B_2| = |B_3| = \frac{|B_x - B_y|}{\sqrt{3}},$$

So a field lying in the diamond xy plane at a general angle (neither along [110] nor $[1\bar{1}0]$, i.e. $B_x \neq \pm B_y$, $B_x, B_y \neq 0$) gives two different pairs: four dips.

Is it possible with the present set-up? No. The coils give lab fields only along Y and Z, and the diamond is fixed at 45°, so every Y field has $B_x = B_y$ (Eq. 4), i.e. $|B_2| = |B_3| = 0$ at $B_z = 0$, which is Case 3. The configuration would require either a third coil along lab X (giving $B_x \neq B_y$) or a rotation of the PCB about the vertical axis to an angle different from 45° (together with $B_z = 0$). By contrast, the standard four-dip case *can* be changed within the set-up: reversing the sign of one coil current gives $B_x = B_y = -B_z$, i.e. **B** along $-NV_4$, which is a Case-4 spectrum with NV_4 as the strongly split orientation.

7 Discussion

Comparison of the measured splittings with the coil calculation

In both single-coil series the splitting grows linearly with current (Fig. 10). Through Eq. 2, the slopes give effective coil constants:

Series B: $\Delta\nu = (17.76 \pm 0.04)\, I + 2.19$ MHz $(r = 0.99999)$

$\Delta\nu = (18.35 \pm 0.14)\, I$ MHz (through origin)

$$k_Z^{\text{exp}} = \frac{\sqrt{3}}{2\gamma}\frac{d\Delta\nu}{dI} = \frac{1.7321}{56} \times 18.35 = \mathbf{0.568 \pm 0.004\ mT/A}$$

Series A: $\Delta\nu = (8.16 \pm 0.11)\, I - 1.67$ MHz $(r = 0.99990)$

$\Delta\nu = (7.76 \pm 0.06)\, I$ MHz (through origin)

$$k_Y^{\text{exp}} = \frac{\sqrt{3/2}}{2\gamma}\frac{d\Delta\nu}{dI} = \frac{1.2247}{56} \times 7.76 = \mathbf{0.170 \pm 0.001\ mT/A}$$

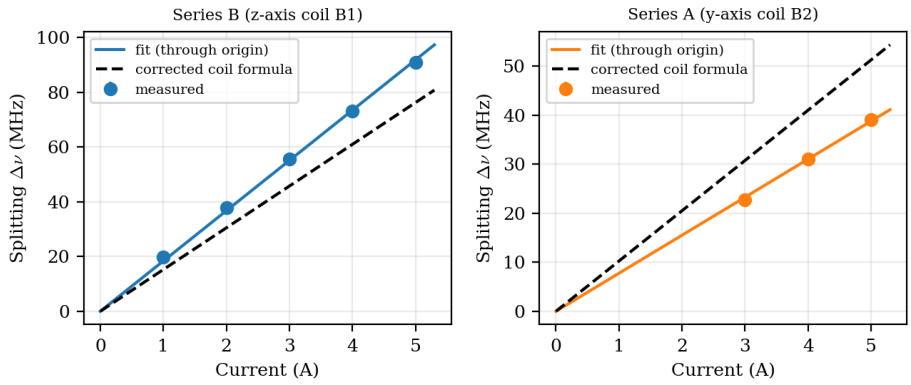


Figure 10: Splitting of the observed dip pair versus current: data with fit errors, linear fit through the origin, and the prediction of the corrected coil formula (nominal geometry).

Relative to the nominal values $k_Z = 0.4712$ and $k_Y = 0.2244$ mT/A, the z-axis coil gives a field larger by a factor 1.20 (+20 %), and the y-axis coil a field smaller by a factor 0.76 (−24 %). The *shape* of the response (linear in I, ratio of the pair splittings $\Delta\nu_\mathrm{A} \propto \sqrt{2/3}\,B_Y$ against $\Delta\nu_\mathrm{B} \propto B_Z/\sqrt{3}$) is as predicted, but the absolute calibration differs from the nominal one. Possible reasons (none of them tested here):

- k_Y depends strongly on distance: $d(\ln B)/d(z) = -3z/(R^2+z^2) = -7.3$ % per cm at $z = 160$ mm. The measured k_Y would correspond to $z \approx 198$ mm, not 160 mm.
- The z-coil value is higher than the *maximum* that a loop of radius 200 mm can give at $z = 0$. It would correspond to a mean winding radius of ≈ 166 mm; the stated 400 mm may be an outer rather than a mean winding diameter, and the finite cross-section of the winding is neglected.
- Small tilts of the diamond/PCB or an error in the nominal 45° mix B_x, B_y, B_z and shift the projections; stray or residual fields offset B; microwave heating shifts D.
- The intercepts of the linear fits (+2.2 MHz, −1.7 MHz) reflect the limited resolution of overlapping dips at the lowest currents.

The full six-dip spectrum at $(4, 4)$ A gives $k_Z^\mathrm{rec} = 0.576$ and $k_Y^\mathrm{rec} = 0.150$ mT/A, which agree in size (k_Z within 2%) with the single-coil values, while k_Y^rec is 0.89 times k_Y^exp. This is a consistency check of the whole analysis, and shows that the discrepancy with the nominal coil formula is systematic and not caused by the fits. For Case 4, the experimental constants require $I_Y/I_Z = \sqrt{2}k_Z^\mathrm{exp}/k_Y^\mathrm{exp} = 4.73$ instead of 2.97, so $(1, 5)$ A is close to the ideal Case 4, as observed.

Other observations:

- The centre of the Series B dip pairs is 2.8716 GHz, close to $D = 2.870$ GHz.
- Dips are ≈ 6–10 MHz wide (FWHM) and widen slightly with the field (inhomogeneity of the applied field over the focal volume and the diamond).
- In the series C runs with B_Z dominant, $(5, 1)$, $(1.4, 1)$ and $(1, 1.4)$ A, the small B_Y shifts the $\mathrm{NV}_1/\mathrm{NV}_4$ dips by only a few MHz around those of $\mathrm{NV}_2/\mathrm{NV}_3$ and they merge into two broadened dips; the apparent splitting is the mean value (marked †). Theory and experiment cannot be compared dip by dip in these runs.

8 Q5. NV centres: insight into physics and applications

Physics. The NV centre is a controllable, room-temperature single quantum system in a solid: it allows the test of spin physics in crystals (spin–orbit and spin–spin interactions, strain and electric-field coupling, hyperfine interaction with $^{14}\mathrm{N}/^{13}\mathrm{C}$), coherent control

(Rabi oscillations, Ramsey and spin-echo experiments), studies of decoherence and of nuclear-spin baths, and quantum entanglement and optical interfaces between spins and photons. As a sensor of local magnetic fields it can image magnetism, superconductivity (vortices, Meissner screening), 2D magnets and current flow in materials with nanometre resolution.

Applications. Vector magnetometry (the four orientations used here give the full field vector), nanoscale NMR/MRI of small samples, magnetic imaging of microelectronics and failure analysis, biomagnetism (magnetotactic bacteria, neuronal activity), nanoscale thermometry and electrometry, rotation sensing, geological and paleomagnetic samples, and building blocks of quantum networks and quantum memories.

9 Conclusions

- ODMR spectra with 1, 2, 3, 4 and 6 dips were observed, in accordance with the symmetry arguments: two dips for $\mathbf{B} \parallel z$ (Series B), three for $\mathbf{B} \parallel [110]$ (Series A), four for the Case-4 configuration with $(I_Z, I_Y) = (1, 5)$ A, and six for $(4, 4)$ A.
- Zeeman splittings from Lorentzian fits grow linearly with current; $B_i = \Delta\nu/2\gamma$ were evaluated for all runs and all calculations are written out in Sec. 6.
- With Eq. 6; $z = 0$ for the z-coil, $z = 160$ mm for the y-coil, $k_Z = 0.4712$ and $k_Y = 0.2244$ mT/A. The experimental constants are 0.568 ± 0.004 and 0.170 ± 0.001 mT/A (z-coil +20 %, y-coil −24 %).
- The six-dip spectrum allowed a complete field-vector reconstruction from the data and confirmed the model $B_1 + B_4 = 2B_2$.

A Fit parameters

Run	dip	f_0 (GHz)	σ_{f_0} (MHz)	FWHM (MHz)	contrast (%)
B1_0_1	1	2.8678	0.06	6.5	3.86
	2	2.8735	0.06	6.0	3.74
B1_0_2	1	2.8673	0.25	10.1	2.68
	2	2.8738	0.21	8.8	2.64
B1_0_3	1	2.8592	0.12	7.0	1.02
	2	2.8677	0.07	6.3	2.46
	3	2.8735	0.06	6.4	2.72
	4	2.8820	0.13	6.7	0.90
B1_0_4	1	2.8551	0.09	7.1	0.96
	2	2.8675	0.05	6.8	2.46
	3	2.8741	0.05	6.6	2.56
	4	2.8862	0.09	6.2	0.89
B1_0_5	1	2.8512	0.10	6.5	1.03
	2	2.8672	0.06	6.9	2.42
	3	2.8745	0.05	6.6	2.67
	4	2.8902	0.11	6.3	0.90
B2_0_1	1	2.8608	0.05	6.9	3.84
	2	2.8806	0.05	6.9	3.63
B2_0_2	1	2.8521	0.05	7.2	3.90
	2	2.8900	0.05	7.0	3.69
B2_0_3	1	2.8437	0.05	7.5	4.00
	2	2.8993	0.05	7.3	3.68
B2_0_4	1	2.8355	0.05	8.1	3.84
	2	2.9087	0.05	7.7	3.58
B2_0_5	1	2.8273	0.05	8.6	3.61
	2	2.9183	0.06	8.0	3.38
B1_B2_4	1	2.8207	0.22	6.1	0.44
	2	2.8350	0.04	8.6	2.71
	3	2.8494	0.23	5.6	0.39

Run	dip	f_0 (GHz)	σ_{f_0} (MHz)	FWHM (MHz)	contrast (%)
	4	2.8964	0.24	5.6	0.38
	5	2.9095	0.05	8.2	2.52
	6	2.9228	0.21	5.5	0.44
B1_B2_5_1	1	2.8269	0.05	7.4	4.00
	2	2.9186	0.05	7.3	3.64
B1_B2_1_5	1	2.8410	0.40	6.1	0.51
	2	2.8604	0.11	10.0	2.38
	3	2.8817	0.11	9.3	2.33
	4	2.9002	0.43	4.9	0.42
B1_B2_1_4_1	1	2.8569	0.05	7.7	3.58
	2	2.8848	0.06	7.5	3.35
B1_B2_1_1_4	1	2.8599	0.07	8.8	3.09
	2	2.8815	0.07	8.4	2.96

Table 11: Lorentzian-fit parameters of all dips (σ_{f_0} = fit standard error).

B Python code used for the fits and field calculation

Please find the code here. The script reads the exported spectra, fits the dips, evaluates the coil formula, rotates the field into the diamond frame, projects it on the NV axes, applies the Zeeman formula and produces the figures.

References

[1] T. Staudacher, *Nuclear magnetic resonance spectroscopy on a nanoscopic sample volume*, Verlag Dr. Hut, 2015.

[2] A. Gruber *et al.*, Science **276**, 2012 (1997), doi:10.1126/science.276.5321.2012.

[3] L. Rondin *et al.*, Rep. Prog. Phys. **77**, 056503 (2014), doi:10.1088/0034-4885/77/5/056503.

[4] R. Schirhagl *et al.*, Annu. Rev. Phys. Chem. **65** (2014), doi:10.1146/annurev-physchem-040513-103659.

[5] A. Dréau *et al.*, Phys. Rev. B **84**, 195204 (2011), doi:10.1103/PhysRevB.84.195204.

[6] L. M. Pham *et al.*, Phys. Rev. B **86**, 121202 (2012), doi:10.1103/PhysRevB.86.121202.

[7] D. Suter, F. Jelezko, Prog. Nucl. Magn. Reson. Spectrosc. **99**, 50–62 (2017).